

Prospect Theory and Time-Varying Regime Transitions in the Cryptocurrency Market: A Two-State Markov Switching Approach

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Abstract

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Keywords: prospect theory, Bitcoin (BTC), Ethereum (ETH), Solana (SOL), TVTP-MRS

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Abstract

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1. Introduction

Traditional asset pricing theories, including the Capital Asset Pricing Model (CAPM) and multifactor pricing frameworks, are grounded in assumptions of rational investor behavior and market efficiency. However, a growing body of evidence in behavioral finance suggests that these assumptions are frequently violated, particularly in highly volatile, sentiment-driven markets. Investor psychology, rather than purely fundamental information, often plays a central role in shaping asset prices, return dynamics, and volatility patterns. In this context, prospect theory (PT), pioneered by Kahneman and Tversky (1979), has become one of the most influential frameworks for explaining decision-making under risk and uncertainty. By emphasizing reference dependence, loss aversion, and probability weighting, PT provides a psychologically realistic basis for understanding why investors may systematically overreact or underreact to gains and losses, thereby generating anomalies that standard rational models struggle to explain.

The relevance of PT is particularly pronounced in speculative financial environments in which valuation anchors are weak, and investor sentiment changes rapidly. Among such environments, the cryptocurrency market stands out as an important empirical setting. Compared with traditional equity markets, cryptocurrencies are characterized by continuous trading, extreme volatility, limited fundamental valuation benchmarks, strong narrative sensitivity, and a high prevalence

of retail-driven participation (Panagiotidis et al., 2019). These features make cryptocurrency markets particularly susceptible to behavioral influences, including overconfidence, extrapolative beliefs, loss aversion, and sharp sentiment reversals during periods of market stress (Almeida et al., 2023; Petkova et al., 2023; Kulbhaskar et al., 2023). As a result, PT offers a promising framework for explaining not only return anomalies in cryptocurrencies but also the cyclical shifts in market sentiment that often underlie prolonged booms and abrupt crashes.

Although previous studies have widely recognized the explanatory power of PT in behavioral finance (Barberis, 2013; Gregoriou et al., 2019), most of this literature has focused on cross-sectional pricing implications or unidirectional return predictability. Much less attention has been paid to how PT-related behavioral signals contribute to the dynamic evolution of market states over time. Yet financial markets, and especially cryptocurrency markets, frequently exhibit pronounced cyclical patterns, most visibly in the alternation between bull and bear phases. These transitions are unlikely to be fully captured by constant-expectation models. The Markov regime-switching (MRS) model proposed by Hamilton (1989) provides a natural framework for modelling such latent market states. More recently, time-varying transition probability Markov regime-switching (TVTP-MRS) models have extended this approach by allowing exogenous variables to influence the probability of remaining in a market regime or switching between regimes. Tan et al. (2021), for example, incorporated Google Trends data into a TVTP-MRS model to capture sentiment-driven volatility dynamics in the BTC market, demonstrating that behavioral variables can enhance the interpretation of market-state changes.

This study addresses the gap by introducing a quantified PTV factor into a two-state TVTP-MRS model to examine its impact on regime-switching dynamics in the cryptocurrency market. Using daily data to construct PTV and weekly data for regime estimation, the study focuses on three major cryptocurrencies, Bitcoin (BTC), Ethereum (ETH), and Solana (SOL), over their available sample periods ending on 5 September 2025. The research investigates whether PTV significantly influences transition probabilities between bull and bear market states, whether its effects differ across transition directions, and whether its behavioral role varies across cryptocurrencies with distinct market characteristics. By doing so, this study extends the PT literature from static return predictability to dynamic regime evolution. It offers new evidence on the behavioral foundations of cyclical risk in digital asset markets.

This study makes three main contributions. First, it extends the behavioral asset-pricing literature by integrating PT into a dynamic market-state framework. While prior studies have shown that investor psychology affects asset returns, relatively few have explained how PTV influences the timing, persistence, and reversal of bull and bear regimes. By treating PTV as a behavioral conditioning variable in the transition probability matrix, this study provides direct evidence that psychologically perceived return distributions shape market regime dynamics.

Second, this study contributes to the growing literature on cryptocurrency market behavior by highlighting heterogeneity across major digital assets. BTC, ETH, and SOL differ markedly in terms of market maturity, ecosystem structure, narrative dependence, and investor participation. BTC is often viewed as the most macro-sensitive and institutionally visible cryptocurrency. In contrast, ETH is more closely tied to platform-based valuation narratives and innovation cycles, and SOL exhibits a shorter but more event-sensitive market history. Examining these assets within a unified TVTP-MRS framework enables this study to determine whether the impact of PTV is broadly consistent across the cryptocurrency market or asset-specific.

Third, this study provides a novel empirical framework for assessing cyclical behavioral risk in highly speculative markets. The proposed methodology quantifies PTV using cumulative PT. It embeds it within a probabilistic regime-switching framework, offering a replicable approach for future research on behavioral state dependence in digital assets and other volatile markets. This approach not only enriches the empirical toolkit for studying cryptocurrency dynamics but also has practical implications for market monitoring, risk management, and early-warning systems that seek to detect behaviorally driven regime shifts before they fully materialize.

The remainder of this study is structured as follows. Section 2 reviews the relevant literature and develops the hypotheses. Section 3 describes the data, the construction of PTV, and the model specification. Section 4 presents the empirical results, including the regime-switching estimates, event-aligned validation, and robustness tests. Section 5 discusses the study's practical implications and limitations. Finally, Section 6 concludes.

2. Literature Review

PT, introduced by Kahneman and Tversky (1979) and later refined into the cumulative form by Tversky and Kahneman (1992), offers a psychologically grounded alternative to the expected utility framework, emphasising reference dependence, loss aversion, and probability weighting. These behavioral elements fundamentally reshape how investors perceive risk and make decisions under uncertainty. In financial economics, PT has been widely recognized for its ability to explain anomalies that traditional rational models fail to capture, such as asymmetric return reactions and excess volatility. Barberis and Thaler (2003) outlined how PT principles can be embedded in asset pricing models, and, more recently, Barberis et al. (2016) proposed a method for empirically quantifying investor sentiment using a metric termed PTV. Their findings indicated that assets with more negatively perceived return distributions (i.e., lower PTV) tend to yield higher subsequent returns, consistent with the notion of loss-averse investors demanding greater compensation for holding such assets.

Despite growing interest in behavioral finance, much of the literature has focused on cross-sectional return predictability, with limited exploration of how behavioral

factors contribute to the dynamic evolution of market states. Financial markets often exhibit clear cyclical patterns, most notably the alternation between bull and bear regimes, which are not easily explained by constant rational expectations. Hamilton (1989) addressed this gap by introducing the MRS model to capture latent structural shifts in macroeconomic and financial time series. Subsequent applications in finance have demonstrated the efficacy of this approach in identifying regime-dependent asset return dynamics (Ang & Timmermann, 2012; Guidolin & Timmermann, 2007). However, traditional MRS models typically assume constant transition probabilities between regimes, failing to account for the role of time-varying behavioral inputs in driving market cycles.

Recent empirical studies have expanded the application of regime-switching models by incorporating TVTP structures, in which exogenous variables influence the probability of switching between regimes (Tan et al., 2021). These models have been employed not only to analyze state-dependent return dynamics and volatility patterns, but also to improve forecasting accuracy and scenario analysis across various financial and commodity markets. Smith and Layton (2007) were among the first to demonstrate the forecasting advantages of TVTP-MRS models in the context of business cycle analysis, showing that incorporating leading indicators into the transition function improved alignment with National Bureau of Economic Research (NBER)-defined economic regimes by 24%. In a different context, Alola (2021) investigated the regime-dependent impact of crude oil prices on Saudi Arabia's real estate market, revealing that oil price shocks exerted greater influence during high-volatility regimes, with global risk factors such as the Volatility Index (VIX) playing a key role in state transitions. Nonetheless, few studies have explicitly operationalized PT within a dynamic market state model. While indicators such as media tone, investor attention, and volatility indexes have been incorporated into regime-switching frameworks, PT's cumulative value function remains underutilized as a behavioral signal that can dynamically influence market regime probabilities. This study aims to fill this gap by incorporating the PTV as a predictor of time-varying transition probabilities in a TVTP-MRS framework. By doing so, it bridges a critical theoretical and empirical disconnect between behavioral asset pricing and dynamic financial market modelling.

The cryptocurrency market provides a particularly suitable empirical setting for testing behaviorally enriched regime-switching models. Compared with traditional equity markets, cryptocurrencies are more heavily influenced by retail trading, speculative narratives, investor attention, and rapid belief updating. Using retail-level trading data, Kogan et al. (2024) show that the same investors who behave contrarily in stocks and gold tend to follow a momentum-like strategy in cryptocurrencies, suggesting that crypto trading is shaped by a distinct expectation-formation process. Relatedly, Al Guindy (2021) finds that investor attention significantly increases cryptocurrency price volatility, while Koutmos (2023) documents a robust positive relationship between investor sentiment and BTC price changes. Aysan et al. (2023) further show that BTC-specific sentiment positively co-moves with returns and has

predictive content, especially over shorter horizons. From a PT perspective, Chen et al. (2022) provide direct evidence that cryptocurrencies with higher PTV tend to earn lower future returns, implying that psychologically attractive return distributions can induce relative overpricing in this market. Taken together, these findings suggest that cryptocurrency prices are shaped not only by fundamentals or macro-financial conditions, but also by investor psychology, sentiment fluctuations, and state-dependent belief formation.

At the same time, the cryptocurrency market exhibits pronounced regime dependence, making it a natural environment for TVTP-MRS models. Bouri et al. (2022) show that cryptocurrency returns co-move in a state-dependent manner, with predictive relationships differing across bullish and bearish regimes. More recent evidence indicates that market-state transitions are not merely statistical artifacts but are closely tied to shifts in investor beliefs and trading behavior. Hsieh et al. (2025), for example, demonstrate that momentum profitability in cryptocurrencies depends critically on state transitions, which arise primarily in persistent up-up regimes and weaken under other transition patterns. In addition, recent Markov-switching applications to the cryptocurrency market show that allowing transition probabilities to vary with external or market-based signals improves the modelling of nonlinear volatility dynamics and regime changes (Zhu et al., 2025).

Nevertheless, the existing literature still provides limited insight into how a PTV measure enters the regime-transition mechanism itself, rather than merely explaining contemporaneous returns or volatility. This omission is particularly important in cryptocurrency markets, where the role of investor sentiment is unlikely to be uniform across market states. The behavioral meaning of a change in PTV may differ depending on whether the market is already in a bull or bear regime. In some cases, an increase in PTV may reinforce bullish sentiment and strengthen bull-state persistence; in other cases, it may reflect speculative overheating and coincide with greater regime instability. Likewise, a decline in PTV may not simply accompany falling prices but may also alter the persistence of bear regimes or the probability of transition out of fragile expansion phases. Existing studies, therefore, leave unresolved a central question: how does a quantified PTV affect the probabilities of remaining in, or switching between, bull and bear market states, and does this effect vary across major cryptocurrencies?

This study addresses that question by embedding PTV directly into a two-state TVTP-MRS framework for BTC, ETH, and SOL. Rather than treating investor sentiment as a general pricing correlate, the analysis examines how the behavioral content of PTV enters the transition equation and affects state-dependent regime dynamics. In particular, the study examines whether changes in PTV strengthen or weaken bull-state persistence, whether they reinforce or erode bear-state persistence, and whether these effects differ systematically across cryptocurrencies with different market structures, narrative dependence, and investor bases. By doing so, this study

extends the literature from static behavioral pricing to the dynamic evolution of market states. It provides a more precise explanation of how quantified investor sentiment shapes cyclical risk in cryptocurrency markets.

2.2 Hypothesis Development

PTV in financial research lies not only in its critique of the expected utility framework, but also in its ability to translate psychologically distorted perceptions of gains and losses into empirically testable implications. By emphasizing reference dependence, loss aversion, and probability weighting, PT suggests that investors do not respond to return distributions in an objectively linear manner; rather, they evaluate outcomes through a psychologically filtered lens (Kahneman & Tversky, 1979; Tversky & Kahneman, 1992). This insight becomes especially important once PT preferences are operationalized through a quantified measure such as PTV. In that form, the theory moves beyond a general behavioral intuition and becomes a concrete valuation variable that reflects how investors subjectively perceive the attractiveness or undesirability of return distributions. Prior studies have shown that PTV has explanatory power for expected returns and asset pricing anomalies, indicating that investor behavior is shaped not merely by realized returns themselves, but by how those returns are psychologically encoded (Barberis et al., 2001; Barberis et al., 2016; Gregoriou et al., 2019).

This perspective is particularly relevant in cryptocurrency markets, where valuation anchors are weak, speculative narratives are strong, and investor beliefs are revised rapidly. Under such conditions, the importance of PTV is not simply that it captures sentiment in a broad sense, but that it links sentiment to a structured behavioral mechanism grounded in PT. Recent evidence further shows that sentiment-related forces affect cryptocurrency dynamics well beyond contemporaneous pricing. Online attention and sentiment influence cross-cryptocurrency correlations, economic sentiment shapes connectedness across major crypto assets, and sentiment-enhanced models outperform purely price-based specifications in BTC forecasting (Aslanidis et al., 2024; Hájek et al., 2023; Osman et al., 2024). If behavioral variables can alter market dependence, predictive structure, and broader dynamics, it is reasonable to expect that a PTV measure may also enter the transition mechanism itself. Within a TVTP-MRS framework, this implies that PTV may affect the probability that the market remains in, or switches between, bull and bear regimes. Accordingly, the first hypothesis is stated as follows:

Hypothesis 1: PTV significantly influences the transition probabilities between bull and bear regimes in cryptocurrency markets.

At the same time, there is little reason to expect the effect of PTV to be directionally uniform. The behavioral logic of PT itself is asymmetric: investors do not react to favorable and unfavorable return perceptions as mirror images of one another. This asymmetry becomes even more relevant in cryptocurrency markets, where sentiment shocks are often amplified by abrupt repricing, jump behavior, and nonlinear

market adjustment. Evidence suggests that sentiment plays a meaningful role in the formation and transmission of jumps in cryptocurrency markets, implying that behavioral forces are capable of affecting market conditions in a state-dependent rather than purely symmetric manner (Aysan et al., 2024). More broadly, regime-switching research has long established that persistence and transition mechanisms may differ fundamentally across expansionary and contractionary phases (Hamilton, 1989; Guidolin & Timmermann, 2007; Ang & Timmermann, 2012). For this reason, an increase in PTV need not have the same implication for the persistence of a bull regime as for the persistence of a bear regime. It may reinforce optimism and prolong favorable states in some circumstances, yet in others it may coincide with overheating, instability, or changes in the likelihood of regime reversal. It motivates a second hypothesis centered on directional asymmetry in the transition mechanism:

Hypothesis 2: The effect of PTV on regime transitions is asymmetric across transition directions, such that its influence differs between bull-state persistence and bear-state persistence.

A further issue concerns whether the transition effect of PTV is likely to be stable across cryptocurrencies. Although BTC, ETH, and SOL are all major digital assets, they differ substantially in market maturity, ecosystem structure, investor composition, and narrative dependence. These differences matter because PTV is not a generic sentiment index detached from asset characteristics; rather, it is a valuation measure derived from the perceived attractiveness of return distributions. As a result, the way in which PTV enters regime dynamics may depend on how investors interpret the return environment of each asset. Recent evidence supports this expectation. Sentiment has been shown to affect herding behavior differently across cryptocurrency groups, while the relationship between cryptocurrencies and broader macro-financial uncertainty also varies meaningfully across assets (Chen & Nguyen, 2024; Civelli & Jackson, 2025). Such findings imply that the same behavioral valuation signal need not translate into identical transition effects across BTC, ETH, and SOL. Instead, differences in market structure and investor participation may generate heterogeneous responses of regime persistence and switching probabilities to changes in PTV. Therefore, the third hypothesis is proposed as follows:

Hypothesis 3: The impact of PTV on regime-switching dynamics differs across major cryptocurrencies, reflecting heterogeneity in market structure, investor participation, and narrative dependence.

3. Data Preparation and Model Construction

This study employs daily price and returns data for three major cryptocurrencies, BTC, ETH, and SOL, with a common sample end date of 5 September 2025. The BTC sample spans 2 June 2013 to 5 September 2025, the ETH sample spans 11 May 2016 to 5 September 2025, and the SOL sample spans 1 May 2021 to 5 September 2025. Daily data are used for return construction and PTV computation, and the regime-switching

estimation is conducted at the weekly frequency. Daily closing-price data for BTC, ETH, and SOL were obtained from Investing.com. The three series were collected from the same provider to ensure consistency in variable definitions, timestamp conventions, and sample coverage. Because cryptocurrencies trade continuously on a 24-hour, 7-day basis, the observations were aligned using a common daily cutoff convention before the construction of daily log returns and subsequent weekly aggregation. Because cryptocurrencies trade continuously on a 24-hour, 7-day basis, the observations were aligned according to the common daily timestamps reported by the data provider to avoid artificial discontinuities arising from heterogeneous exchange closing conventions and time zone differences.

Daily returns are computed as log returns, $r_t = \ln(P_t/P_{t-1})$, where P_t denotes the daily closing price at the chosen cut off time. To construct excess returns for the PTV calculation, the study uses the U.S. 10-year Treasury yield as the risk-free benchmark. The yield series is converted into a daily rate using a standard day-count convention, and the daily risk-free-adjusted return is defined as $R_t = r_t - r_{ft}$. Since daily risk-free rates are small relative to cryptocurrency volatility, the empirical analyses additionally verify robustness to alternative excess return definitions, including setting $r_{ft} = 0$, to ensure that the choice of the risk-free proxy does not mechanically drive the results.

A transparent, replicable data-cleaning protocol is implemented before model estimation. First, the raw series are screened for missing values in the price and timestamp columns. Observations with missing closing prices are removed, and gaps are not imputed to avoid generating mechanically induced returns. Second, duplicated observations at identical timestamps are resolved to maintain a unique daily record per asset by retaining a single observation and discarding remaining duplicates. Third, to mitigate the influence of extreme observations while preserving the informational content of the return series, daily returns are winsorized at the 0.1 per cent level in both tails. Specifically, returns below the 0.1th percentile are set equal to the 0.1th percentile, and returns above the 99.9th percentile are set equal to the 99.9th percentile, using the full sample distribution for each cryptocurrency. This treatment reduces sensitivity to sporadic data glitches and microstructure-related extremes without deleting observations. Fourth, all series are sorted and aligned to a strictly ordered daily calendar with no repeated dates. When analyses require joint estimation across cryptocurrencies, the sample is restricted to the intersection of available trading days across the relevant assets; otherwise, each cryptocurrency is analyzed on its maximum available sample to retain information.

In the empirical analysis, the regime-switching models are estimated at the weekly frequency to improve regime interpretability in highly volatile cryptocurrency markets. To ensure consistency across the descriptive statistics, correlation analysis, and subsequent model estimation, all variables used in the TVTP-MRS and MRS models are aligned to weekly observations. Specifically, daily risk-free-adjusted log returns are

aggregated by summing all available daily observations within each calendar week, whereas weekly PTV is calculated as the arithmetic mean of the available daily PTV values within the same week. The summation preserves the time-additive property of log returns. Unlike returns, PTV is a behavioral valuation indicator rather than an additive quantity across time. Therefore, averaging the available daily PTV observations provides a representative measure of the behavioral valuation environment prevailing during each week, rather than a cumulative behavioral measure, while also mitigating high-frequency fluctuations in the daily PTV series.

3.2 Prospect Theory Value under the Cumulative Prospect Theory Framework

3.2.1 Construction of PTV

Following the cumulative PT framework of Tversky and Kahneman (1992), this study quantifies PTV as a market-based measure of investors' subjective valuation of a cryptocurrency's return distribution. In the PTV framework, the traditional utility function is replaced by a value function defined over gains and losses relative to a reference point, and objective probabilities are transformed into decision weights through a probability weighting function.

To define notation, for each cryptocurrency and each date t , we construct an empirical return distribution using a rolling window of the past 60 days' risk-free adjusted returns, $\{R_{t-N}, \dots, R_{t-1}\}$, where $R_t = r_t - r_{ft}$. The returns in the window are sorted in ascending order and denoted by $R_{(i)t}$, $i = -m, \dots, n$, where negative outcomes and positive outcomes are indexed separately, with $R_{(i)t} \leq R_{(j)t}$ for $i < j$. Consistent with an empirical distribution representation, each ordered outcome is assigned an equal probability mass, $p_i = 1/(m + n + 1)$. This empirical distribution is constructed separately for each cryptocurrency. The PTV at date t is then computed as the probability-weighted sum of the value function evaluated at the ordered outcomes,

$$PTV_t = \sum_{i=-m}^n \pi_i v(R_{(i)t}), \quad (1)$$

where $R_{(i)t}$ denote the i^{th} order of risk-free adjusted rate of return, arranged in ascending order. For $i < 0$, $R_{(i)t}$ is a negative value, and for $i > 0$, $R_{(i)t}$ is a positive value. Accordingly, the resulting PTV_t series is used as a time-varying behavioral conditioning variable in the regime-switching analysis. Here, the value and probability weighting functions are defined as follows:

$$v(x) = \begin{cases} x^\alpha, & x > 0 \\ -\lambda(-x)^\alpha, & x < 0 \end{cases} \quad (2)$$

$$(3)$$

$$\pi_i = \begin{cases} \omega^+(p_i + \dots + p_n) - \omega^+(p_{i+1} + \dots + p_n), & 0 \leq i \leq n, \\ \omega^-(p_{-m} + \dots + p_i) - \omega^-(p_{-m} + \dots + p_{i-1}), & -m \leq i < 0, \end{cases} \quad (4)$$

$$\omega^+(P) = \frac{P^\gamma}{(P^\gamma + (1-P)^\gamma)^{\frac{1}{\gamma}}}; \quad \omega^-(P) = \frac{P^\delta}{(P^\delta + (1-P)^\delta)^{\frac{1}{\delta}}}.$$

where $\alpha \in (0,1)$, $\gamma \in (0,1)$, $\delta \in (0,1)$, and $\lambda > 1$. Equations (2)-(4) formalize PTV. The value function $v(x)$ captures investors' asymmetric sensitivity to gains and losses around a reference point, characterized by loss aversion (λ) and diminishing sensitivity (α). The decision weights π_i adjust objective probabilities based on cumulative outcomes, reflecting cognitive biases. The weighting functions $\omega^+(P)$ and $\omega^-(P)$ further distort probabilities, overweighting small probabilities and underweighting large ones, governed by curvature parameters γ and δ . Together, these equations quantify investor psychological biases influencing perceived returns. The calibration results of Tversky and Kahneman (1992) are:

$$\alpha = 0.88, \lambda = 2.25, \\ \gamma = 0.61, \delta = 0.69.$$

To facilitate comparability with the standard cumulative PT specification. In additional analyses, we verify that the main conclusions remain robust to alternative parameter choices. We also adopt these parameter values in the study. Figure 1 depicts the trend of the value function and the probability weighting functions proposed by Tversky and Kahneman (1992).

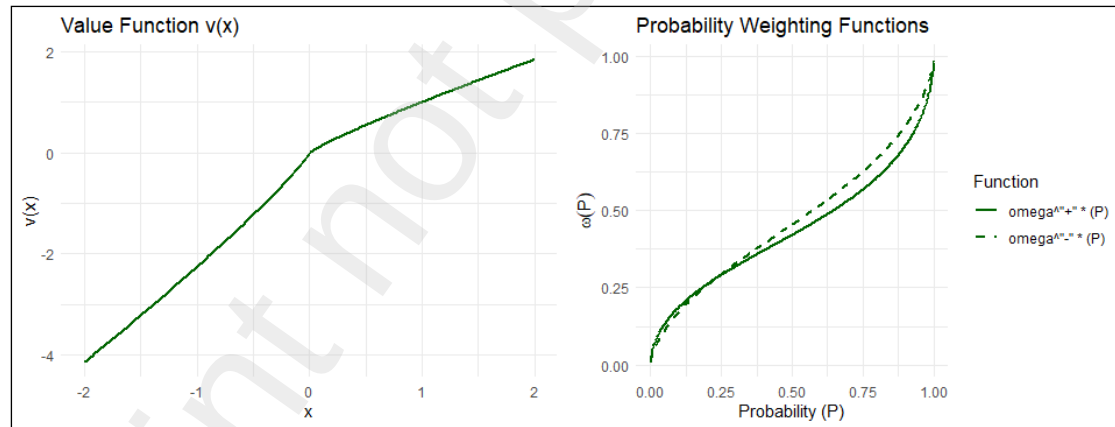


Figure 1. Curves of the variation in the value function and the probability weighting function

In operational terms, PTV_t summarizes the PT attractiveness of the recent return distribution for a given cryptocurrency. A lower PTV_t indicates that recent outcomes are evaluated more negatively under loss aversion and probability distortion, whereas a higher PTV_t reflects a more favorable subjective valuation of recent outcomes. This time-varying PTV series is subsequently used as a state dependent behavioral conditioning variable in the regime-switching analysis.

3.2.2 PTV calculation

This study constructs PTV by closely following the empirical logic of Barberis et al. (2016), while adapting the measurement frequency to the cryptocurrency market. The key principle in Barberis's study is that investors form a mental representation of an asset's recent return distribution, and that this perceived distribution can be mapped into a single valuation metric using cumulative PT. In our setting, the same principle applies, but the cryptocurrency market exhibits substantially faster information incorporation, continuous trading, and more pronounced high-frequency variation than equity markets. Using monthly returns would therefore excessively smooth the return distribution and may mask short-lived swings in perceived gains and losses that are central to regime dynamics. Accordingly, we implement the PTV construction at the daily frequency so that the PT-based valuation measure is aligned with the higher frequency regime switching structure examined in the subsequent TVTP-MRS analysis.

Specifically, for each cryptocurrency and each day t , we use a rolling window of $N = 60$ daily excess returns to represent investors' recent experience set. The 60-day window provides a parsimonious approximation of a short-horizon memory and is long enough to form a non-degenerate empirical distribution while remaining responsive to the rapid evolution of cryptocurrency market conditions. Daily excess returns are defined as $R_t = r_t - r_{ft}$, where r_t is the daily log return and r_{ft} is the daily risk-free rate converted from the U.S. 10-year Treasury yield under a standard day count convention. To mitigate the influence of extreme observations and potential data irregularities that are more prevalent in high volatility markets, we winsorize R_t at the 0.1 per cent level in both tails before constructing the rolling window distribution. The value function and probability weighting functions follow Tversky and Kahneman (1992) using the benchmark parameterization, $\alpha = 0.88$, $\lambda = 2.25$, $\gamma = 0.61$, and $\delta = 0.69$, which facilitates comparability with the standard cumulative PT specification.

After these steps, the 60 daily excess returns in the rolling window are sorted in ascending order and assigned equal probability mass $p_i = 1/60$. The resulting ordered outcomes and their cumulative decision weights are then used to compute PTV_t as in Equations (5) and (6).

$$(R_{(-m)t}, \frac{1}{60}; R_{(-m+1)t}, \frac{1}{60}; \dots, R_{(-1)t}, \frac{1}{60}; R_{(1)t}, \frac{1}{60}; \dots, R_{(n-1)t}, \frac{1}{60}, R_{(n)t}, \frac{1}{60}) \quad (5)$$

Among them, the m risk-free adjusted returns corresponding to $R_{(-m)t}, \dots, R_{(-1)t}$ are negative and n risk-free adjusted returns corresponding to $R_{(1)t}, \dots, R_{(n)t}$ are positive. Then, the PTV of each cryptocurrency at time t is:

$$PTV_t = \sum_{i=-m}^{-1} v(R_{(i)t}) \left[\omega^- \left(\frac{i+m+1}{60} \right) - \omega^- \left(\frac{i+m}{60} \right) \right] + \sum_{i=1}^n v(R_{(i)t}) \left[\omega^+ \left(\frac{n-i+1}{60} \right) - \omega^+ \left(\frac{n-i}{60} \right) \right]. \quad (6)$$

Because the computation requires 60 prior observations, the daily PTV series for each cryptocurrency begins on the 61st trading day of its sample period and ends on 5 September 2025. The PTV measure is initially constructed at the daily frequency using a rolling window of daily excess returns. For weekly model estimation, the available daily PTV values are averaged within each calendar week. The resulting weekly PTV therefore represents the average behavioral valuation environment prevailing during that week, with each daily PTV observation being constructed from its corresponding rolling 60-day return window.

Table 1 presents the interpretation of PTV signs in terms of investor psychological perception, offering critical behavioral context for the time-varying transition probabilities modelled in the TVTP-MRS framework. When PTV is positive, it indicates that investors, after subtracting the risk-free rate, perceive the asset's return profile as psychologically appealing. This reflects a subjective valuation that is above the internalized reference point, typically motivating continued investment or optimistic positioning. Conversely, a negative PTV signifies that the psychological value of the asset is viewed as unfavorable. Investors interpret the return prospect as unsettling or unworthy, which reflects the "loss domain" described in PT. This perceived disutility increases the likelihood of aversive responses such as withdrawal, inaction, or even panic selling, particularly under uncertainty. By incorporating PTV as an exogenous variable in the TVTP, the model captures how shifts in investor sentiment, quantified via PTV, systematically influence the probability of staying in or switching between bull and bear market regimes.

Table 1. PTV symbols explanation

PTV symbols interpretation	
$PTV_t > 0$	Indicates that the asset's return profile at time t (based on historical 60-day returns) is perceived as psychologically attractive, with subjectively weighted gains outweighing losses under the value function.
$PTV_t < 0$	Indicates that the asset at time t (based on historical 60-day returns) is perceived as psychologically unattractive. This reflects a subjective weighting in which perceived losses outweigh gains, leading investors to view the overall prospect as unsettling, emotionally aversive, or not worthwhile relative to their reference point.

3.3 PT and Cryptocurrency Regime Dynamics

The TVTP-MRS model is used to estimate the probability that each week in the effective estimation sample belongs to either the bull or bear regime. It is also employed to examine whether and how PTV affects the probabilities of remaining in or switching between the two market regimes across BTC, ETH, and SOL.

3.3.1 Modelling market cyclicalities

This study uses the TVTP-MRS model to analyze the variability of PT's impact on the cryptocurrency market under different market states, as shown in the model below:

$$R_{t+1} = \mu_{S_t} + \beta_{S_t} R_t + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma_{S_t}^2) \quad (7)$$

where R_t represents the risk-free adjusted rate of return, calculated by summing the available daily risk-free-adjusted log returns within week t ; S_t is a latent state variable representing the market state at time t , the two states are identified statistically during model estimation, without imposing bull or bear labels in advance. Their economic interpretations are subsequently determined by examining the estimated regime-specific characteristics and their correspondence with historical cryptocurrency market conditions. μ_{S_t} is the state-dependent mean return, which varies according to the market state S_t , β_{S_t} represents the state-dependent coefficient on lagged returns, capturing how past returns influence the current return conditional on the regime; ε_t is the random error term, which represents random perturbations, that is, random fluctuations that the model fails to explain. Note that the variance of the error term, $\sigma_{S_t}^2$, is also state-dependent.

Importantly, the two latent regimes are not defined mechanically by the signs or relative magnitudes of their regime-specific conditional means. A higher conditional mean does not, by itself, imply that a state represents a bull regime, just as a lower conditional mean does not automatically identify a bear regime. Instead, the model first estimates two statistically distinct latent states without imposing economic labels. Their economic interpretations are assigned only after estimation by jointly examining regime-specific volatility, the timing and persistence of the smoothed state probabilities, and their correspondence with documented cryptocurrency expansion, contraction, and stress episodes. Accordingly, the terms “bull regime” and “bear regime” used in the subsequent analysis should be understood as ex-post economic interpretations of the estimated latent states rather than classifications based solely on positive or negative conditional mean returns.

3.3.2 Transition probability matrix with time-varying PT factors

This study assumes that the transition probability is time-varying, with regime transition probabilities that vary over time as a function of PTV. This approach enables the dynamic adjustment of the probability of staying in or switching between bull and bear markets, based on behavioral factors.

For the two-state TVTP-MRS model, State 1 and State 2 represent two latent market regimes whose economic meanings are determined after estimation. The time-varying transition probability matrix at time t is defined as:

$$\mathbf{P}_t = \begin{bmatrix} P_{11_t} & P_{12_t} \\ P_{21_t} & P_{22_t} \end{bmatrix}, \quad (8)$$

where

$$P_{ijt} = P(S_{t+1} = j | S_t = i, PTV_t), i, j = 1, 2. \quad (9)$$

Following estimation, the economic meanings of the two latent states are determined ex post using a common identification procedure for BTC, ETH, and SOL. The interpretation jointly considers three complementary features: **(i)** the relative magnitude of regime-specific volatility, **(ii)** the timing and persistence of the smoothed state probabilities, and **(iii)** the correspondence between these probabilities and documented cryptocurrency expansion, contraction, recovery, and stress episodes.

State 1 is characterized by relatively lower volatility and prolonged occupancy during sustained market expansion and recovery periods, whereas State 2 exhibits substantially higher volatility and its smoothed probability tend to increase during major market contraction and stress episodes. The regime-specific conditional means are considered together with these characteristics in interpreting the economic meaning of the estimated states, rather than serving as the sole basis for assigning the regime labels.

Based on this joint evidence, State 1 and State 2 are subsequently referred to as the bull and bear regimes, respectively. These labels therefore represent ex-post economic interpretations of the statistically estimated latent states rather than classifications based solely on the regime-specific conditional means.

The transition probabilities vary over time according to the following logistic functions. By construction, each row of P_t sums to one:

$$P_{11_t} = \frac{e^{a_1 + b_1 \cdot PTV_t}}{1 + e^{a_1 + b_1 \cdot PTV_t}}, \quad (10)$$

$$P_{22_t} = \frac{e^{a_2 + b_2 \cdot PTV_t}}{1 + e^{a_2 + b_2 \cdot PTV_t}}, \quad (11)$$

$$P_{12_t} = 1 - P_{11_t}, \quad (12)$$

$$P_{21_t} = 1 - P_{22_t}. \quad (13)$$

where a_1, b_1, a_2 and b_2 are the parameters to be estimated. Note that P_{11_t} denotes the probability of remaining in State 1; and P_{22_t} denotes the probability of remaining in State 2, following the ex-post regime interpretation, these probabilities are referred to as bull-regime persistence and bear-regime persistence, respectively.

This formulation captures the influence of investor sentiment on the likelihood of staying in or switching between different market regimes.

In this specification, a_1 and a_2 are intercept terms capturing the baseline log-odds of staying in the bull and bear regimes, respectively. The parameters b_1 and b_2 measure how the weekly PTV information available in week t affects the probabilities of remaining in the bull and bear regimes in week $t + 1$, respectively. Equivalently, when the return equation is indexed at $t + 1$, PTV_t constitutes a one-period-lagged transition covariate. A positive value of b_1 implies that an increase in PTV leads to an increase in the probability of staying in a bull market. In contrast, a negative value of b_2 indicates that a negative coefficient implies that lower PTV increases the probability of remaining in a bear regime.

4. Empirical Analysis and Results

4.1 Descriptive statistics and correlation analysis

Unless otherwise stated, descriptive statistics and pairwise correlations are reported at the weekly frequency, consistent with the estimation samples used in the TVTP-MRS and MRS models. Weekly cryptocurrency returns are constructed by summing the available daily risk-free-adjusted log returns within each calendar week, whereas weekly PTV is calculated as the arithmetic mean of the available daily PTV observations within the same week. This consistent aggregation procedure ensures that the statistical properties reported in this section correspond directly to the variables used in the subsequent regime-switching estimation. The variable definitions, descriptive statistics, and unconditional correlations are reported in Tables 2, 3, and 4, respectively.

Table 2. Variable definitions

<i>Variable</i>	<i>Definition</i>
R_t	R_t denotes the cumulative weekly risk-free-adjusted log return of a cryptocurrency in week t , calculated as the sum of the available daily risk-free-adjusted log returns within that week.
PTV_t :	PTV_t denotes the weekly prospect theory value of a cryptocurrency in week t , calculated as the arithmetic mean of the available daily PTV observations within that week.
S_t :	A latent variable represents the state of the market at time t , where $S_t = 1$ represents a bull market, and $S_t = 2$ represents a bear market.

Table 3. Descriptive statistics for cryptocurrencies

Panel A: BTC descriptive statistics					
Variable	Obs	Average	Std. dev	Min	Max
R_t	640	0.0075	0.1103	-0.5532	0.5405
PTV_t	640	-0.0257	0.0172	-0.1007	0.0060
Panel B: ETH descriptive statistics					
Variable	Obs	Average	Std. dev	Min	Max
R_t	486	0.0122	0.1355	-0.5362	0.5673
PTV_t	486	-0.0502	0.0327	-0.1875	0.0332
Panel C: SOL descriptive statistics					
Variable	Obs	Average	Std. dev	Min	Max
R_t	221	0.0070	0.1553	-0.9067	0.4606
PTV_t	221	-0.0621	0.0379	-0.1768	0.0111

Notes: Weekly returns are calculated by summing the available daily risk-free-adjusted log returns within each calendar week, whereas weekly PTV is calculated as the arithmetic mean of the available daily PTV observations. The descriptive statistics are based on the final weekly sample aligned with the TVTP-MRS estimation. Observations unavailable because of the rolling PTV construction and the lag structure of the model are excluded.

Table 4. Unconditional correlation between factors

Panel A: BTC correlation analysis			
Variable	R_{t+1}	R_t	PTV_t
R_{t+1}	1.0000		
R_t	0.0687	1.0000	
PTV_t	0.1214	0.2768	1.0000
Panel B: ETH market correlation analysis			
Variable	R_{t+1}	R_t	PTV_t
R_{t+1}	1.0000		
R_t	0.0864	1.0000	
PTV_t	0.0568	0.3993	1.0000
Panel C: SOL market correlation analysis			
Variable	R_{t+1}	R_t	PTV_t
R_{t+1}	1.0000		

R_t	0.1089	1.0000	
PTV_t	-0.0088	0.3656	1.0000

4.2 Unit root test

Before estimating the TVTP-MRS model, it is necessary to examine the time-series properties of the variables to ensure that the empirical analysis is not affected by non-stationarity. In particular, non-stationary series may lead to spurious inference and undermine the reliability of regime-dependent estimation. To address this issue, this study conducts three complementary unit root tests, namely the Phillips-Perron (PP) test, the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, and the Augmented Dickey-Fuller (ADF) test, for the weekly return series R_t and the PTV series PTV_t of BTC, ETH, and SOL. The results are reported in Table 5.

Table 5. Unit root test for variables

<i>Panel A: Unit root test for BTC</i>						
	PP		KPSS		ADF	
Variables	PP-Statistic	P-Value	KPSS-Statistic	Critical-Value	ADF-Statistic	P-Value
R_t	-695.773***	0.0100	0.0727	0.7390	-6.892***	0.0100
PTV_t	-74.239***	0.0100	0.6022	0.7390	-5.878***	0.0100
<i>Panel B: Unit root test for ETH</i>						
	PP		KPSS		ADF	
Variables	PP-Statistic	P-Value	KPSS-Statistic	Critical-Value	ADF-Statistic	P-Value
R_t	-470.777***	0.0100	0.1883	0.7390	-6.755***	0.0100
PTV_t	-95.349***	0.0100	0.6688	0.7390	-4.775***	0.0100
<i>Panel C: Unit root test for SOL</i>						
	PP		KPSS		ADF	
Variables	PP-Statistic	P-Value	KPSS-Statistic	Critical-Value	ADF-Statistic	P-Value
R_t	-214.922***	0.0100	0.1039	0.7390	-4.959***	0.0100
PTV_t	-49.713***	0.0100	0.7224	0.7390	-3.488**	0.0448

Notes: *, ** and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 5 shows that the weekly return series R_t for BTC, ETH, and SOL are consistently stationary across the three tests. Specifically, both the PP and ADF tests strongly reject the null hypothesis of a unit root, while the KPSS statistics remain below the adopted critical value, indicating that the null of level stationarity cannot be rejected. This provides clear evidence that the return series are stationary and suitable for subsequent regime-switching estimation.

The evidence for the PTV_t series is slightly less sharp than that for returns, but the overall conclusion still supports stationarity. For all three cryptocurrencies, the PP and ADF tests reject the unit-root null, suggesting that the PTV series do not exhibit persistent non-stationary behavior. Although the KPSS statistics for PTV_t , particularly

in the case of SOL, are relatively closer to the critical threshold, they remain below the 1% critical value adopted in this study. Hence, the null of level stationarity is not rejected under this criterion. Taken together, the results indicate that both the return and PTV series are sufficiently stationary for the empirical analysis. Therefore, the subsequent TVTP-MRS estimation is conducted at the level rather than in first differences.

4.3 Baseline TVTP-MRS Results

Before presenting the TVTP-MRS estimates, we benchmark several competing time-series specifications to verify that a regime-switching framework is empirically warranted for BTC weekly returns. Specifically, we compare a linear AR(1) model and an AR(1)-GARCH(1,1) model with two-state MRS models under (i) constant transition probabilities and (ii) time-varying transition probabilities driven by PTV. Table 6 compares the regime-switching specifications with the single-regime AR(1) and AR(1)-GARCH(1,1) benchmarks. Both the constant-transition MRS and TVTP-MRS models provide substantially better fit than the single-regime specifications, supporting the presence of distinct latent market states in BTC weekly returns. The TVTP-MRS model yields a slightly higher log-likelihood than the constant-transition MRS model, indicating a modest improvement in unrestricted in-sample fit. However, the AIC and BIC favor the more parsimonious constant-transition specification after accounting for the additional parameters. Therefore, Table 6 supports the general regime-switching structure, but does not by itself establish the superiority of the TVTP-MRS model. The TVTP-MRS specification is retained as the primary model because it directly corresponds to the study's central research question of whether PTV enters the regime-transition mechanism.

Table 6. Model comparison for BTC weekly returns

Model	Regimes	Transition type	Log-likelihood	AIC	BIC
AR(1)	1	No regime switching	504.77	-1003.55	-990.16
AR(1)-GARCH(1,1)	1	No regime switching	567.97	-1125.94	-1103.63
Two-state MRS	2	Constant	600.05	-1184.10	-1148.41
Two-state TVTP-MRS	2	Time-varying (PTV)	600.93	-1181.86	-1137.25

BTC is used as the benchmark asset for model comparison because it has the longest sample and the most mature market structure. Guided by this evidence, we estimate the two-state TVTP-MRS model for each cryptocurrency and examine the inferred market states. Figures 2-4 plot the filtered and smoothed regime probabilities, which provide an intuitive visualization of the time-varying likelihood of being in bull versus bear conditions. These probability paths also serve as a diagnostic for regime

interpretability and help contextualize the subsequent discussion of state-dependent return dynamics and PTV-driven transition mechanisms.

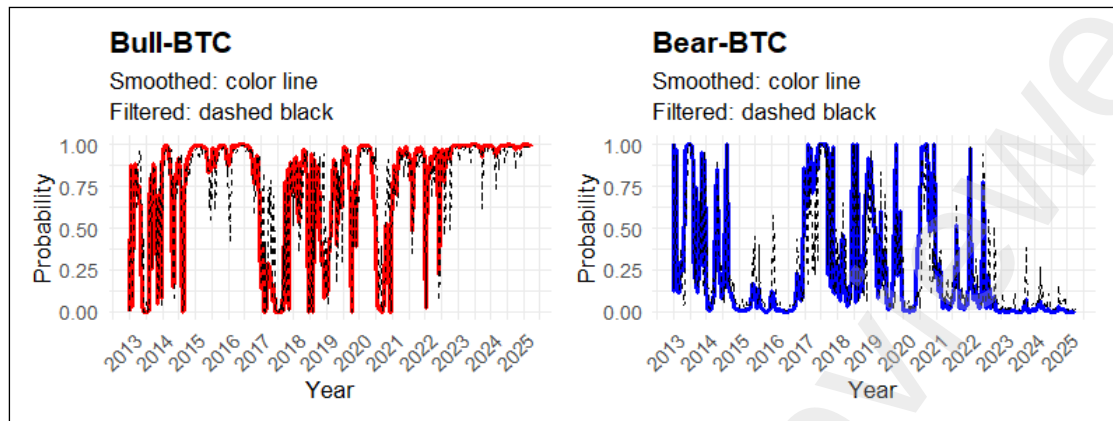


Figure 2. Filter & Smoothed probabilities of weekly BTC (TVTP-MRS)

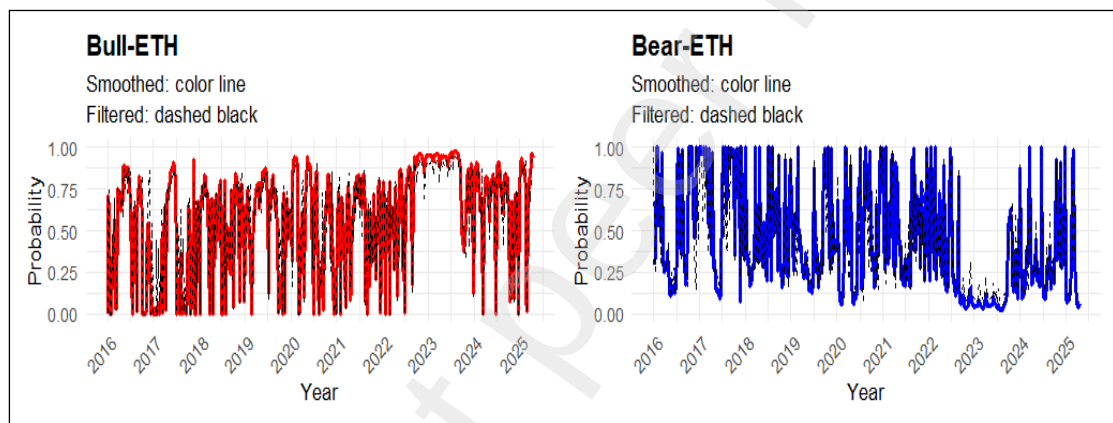


Figure 3. Filter & Smoothed probabilities of weekly ETH (TVTP-MRS)

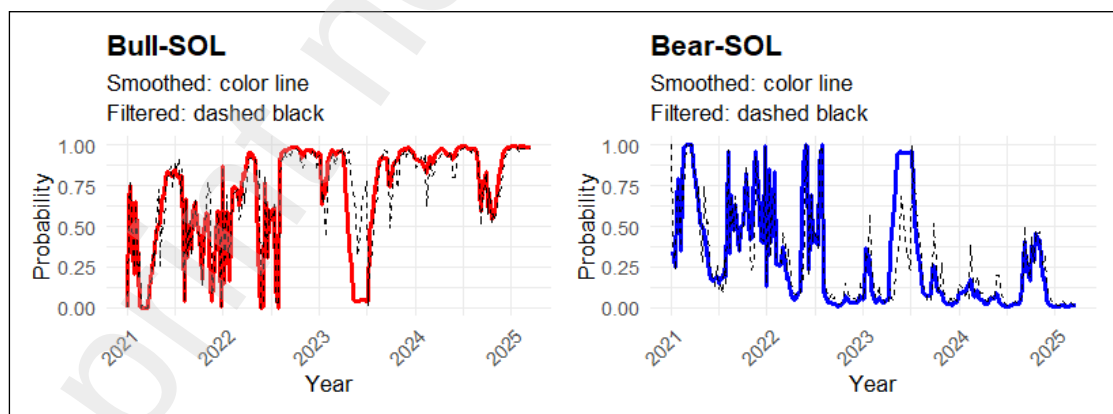


Figure 4. Filter & Smoothed probabilities of weekly SOL (TVTP-MRS)

Building on the model-comparison evidence in Table 6, Figures 2-4 plot the filtered and smoothed regime probabilities from the weekly TVTP-MRS model for BTC, ETH, and SOL. Based on the ex-post economic interpretation of the estimated latent states, a notable pattern across the three assets is a “bull-long, bear-short”

asymmetry: bull-regime probabilities remain close to one for extended intervals, whereas bear-regime probabilities tend to emerge in shorter, episodic bursts. This pattern is visually consistent with widely recognized stress episodes in crypto markets, during which downside risk becomes concentrated, and regime probabilities shift rapidly. For instance, elevated bear-regime probabilities align in timing with the broad 2018 market drawdown, the abrupt global risk-off turbulence in early 2020, and the series of crypto-ecosystem shocks during 2022 that triggered sharp contractions in risk appetite. Conversely, the prolonged dominance of bull-regime probabilities corresponds to sustained expansion phases characterized by persistent inflows and improving sentiment, suggesting that weekly crypto return dynamics are better represented by recurring bull and bear states than by a single-regime process.

Importantly, the figures reveal substantial heterogeneity in regime dynamics across the three cryptocurrencies. Relative to BTC and SOL, ETH exhibits a more “fragmented” probability path, characterized by frequent alternation between high and low regime probabilities and denser probability spikes. This visual contrast does not by itself identify the transition mechanism, but it aligns with the coefficient heterogeneity reported later in Table 7, providing a quantitative basis for interpreting these patterns. BTC, in comparison, shows more persistent regime occupancy, consistent with its greater exposure to macro-liquidity and broad risk-sentiment channels. SOL, while exhibiting clearer regime clustering, has a shorter sample period and experiences pronounced phase shifts after 2021, during which ecosystem-level shocks and recovery episodes can generate concentrated bursts of regime reallocation. Overall, these probability paths suggest that bull phases tend to be more persistent, whereas bear phases are typically abrupt and event-driven, and that ETH in particular is associated with larger and more frequent fluctuations in regime probabilities, reflecting its higher sensitivity to ecosystem-driven valuation cycles and market narratives, such as the 2017 ICO boom and the 2020 DeFi expansion.

Motivated by these regime-probability diagnostics and their alignment with major market episodes, Table 7 next reports the full TVTP-MRS estimation results. Specifically, it summarizes the regime-dependent return dynamics (Panel A), the PTV-linked transition mechanism governing regime persistence and switching (Panel B), providing the quantitative foundation for interpreting the patterns observed in Figures 2-4.

Table 7. TVTP-MRS model results for weekly cryptocurrencies

<i>Panel A: Estimated sample coefficients</i>						
Markets	μ_1	μ_2	β_1	β_2	σ_1	σ_2
BTC	0.006* (0.086)	0.007 (0.623)	0.068* (0.100)	0.220*** (0.010)	0.069*** (0.000)	0.172*** (0.000)
ETH	0.005 (0.360)	0.017 (0.234)	-0.136** (0.045)	0.149* (0.090)	0.066*** (0.000)	0.194*** (0.000)
SOL	-0.001 (0.985)	0.011 (0.727)	-0.210** (0.020)	0.344** (0.049)	0.116*** (0.000)	0.215*** (0.000)
<i>Panel B: Transition probabilities and estimated transition coefficients</i>						
Markets	a_1	a_2	b_1	b_2		
BTC	3.075*** (0.000)	-1.987*** (0.004)	1.175*** (0.010)	-1.036* (0.092)		
ETH	-1.146*** (0.007)	0.714 (0.123)	-0.810* (0.068)	0.426* (0.100)		
SOL	-3.024*** (0.008)	1.216* (0.072)	-2.377* (0.100)	0.725* (0.093)		

Notes: *, **, and *** denote significance at 10%, 5%, and 1% level. The values in parentheses represent p-values.

To complement the quantitative results reported in Table 7, Table 8 provides a behavioral and theoretical interpretation of the estimated transition coefficients for BTC, ETH, and SOL. The table summarizes how the sign and magnitude of weekly PTV influence bull- and bear-state probabilities, linking the numerical coefficients (b_1 and b_2) to observable market dynamics and investor behavior. Positive or negative deviations of PTV from its reference point (zero) are mapped to increases or decreases in regime persistence, highlighting the differential effects across cryptocurrencies. This descriptive summary facilitates intuitive understanding of the directionality and economic plausibility of the TVTP-MRS model coefficients, illustrating how elevated PTV may reinforce or destabilize market regimes depending on the asset and the associated behavioral responses.

Table 8. Behavioral interpretation of regime transitions by PTV sign

PTV Sign and Coefficient Condition	Implication on transition probabilities	Behavioral characteristics	Theoretical explanation
$PTV_t > 0$, $b_1 > 0$, and PTV_t increase (BTC type)	There is an increase in the probability of P_{11_t} $P_{11_t} \uparrow$	Persistence in the bull market	Investors interpret recent return distributions as psychologically attractive, which supports continued risk-taking and strengthens bull-state persistence. In the BTC market, this pattern is consistent with the view that higher PTV reinforces regime stability rather than triggering immediate reversal.
$PTV_t > 0$, $b_1 < 0$, and PTV_t increase (SOL/ETH type)	There is a decrease in the probability of P_{11_t} $P_{11_t} \downarrow$	The bull market is more likely to turn into a bear market	Although return prospects are psychologically attractive, elevated PTV may coincide with cyclical overheating or narrative-driven amplification. In this case, higher PTV is not associated with stronger bull-state stability and may coincide with subsequent regime reallocation.
$PTV_t < 0$, $b_2 > 0$, and PTV_t becomes more negative (SOL/ETH type)	There is a decrease in the probability of P_{22_t} $P_{22_t} \downarrow$	The bear market is more likely to turn into a bull market	As PTV becomes more negative, the estimated probability of remaining in the bear regime declines. This counterintuitive pattern may reflect capitulation or oversold market conditions, under which extreme pessimism may coincide with short covering, bargain buying, or selective re-entry by investors. These responses may increase the probability of transition out of the bear regime.
$PTV_t < 0$, $b_2 < 0$, and PTV_t becomes more negative (BTC type)	There is an increase in the probability of P_{22_t} $P_{22_t} \uparrow$	Persistence in the bear market	When PTV becomes more negative, investors interpret recent outcomes as increasingly unfavorable, which strengthens defensive trading, withdrawal from risky positions, and bearish persistence. In the BTC market, this pattern is consistent with sharper downside reinforcement during major stress episodes.

Note: The interpretations summarized in Table 8 are grouped according to the sign structure of the estimated transition coefficients and therefore apply differently across BTC, ETH, and SOL. The behavioral explanations, including overheating, narrative-driven amplification, capitulation, and oversold-market responses, are economically plausible interpretations of the estimated patterns rather than mechanisms directly identified or separately tested by the empirical model.

4.4 Event-aligned validation of transition mechanisms

To further assess the economic plausibility of the transition coefficients reported in Table 7, we visually align weekly bull-state probabilities with the standardized weekly PTV series and examine their co-movements around major market episodes. This exercise does not introduce new estimation or identification strategies; rather, it serves as a descriptive validation of whether the estimated PTV-linked transition mechanism is directionally consistent with observed regime shifts during salient market events.

For BTC, the bull-state probability path exhibits strong persistence throughout most of the sample (see Figure 5). Bull probabilities remain elevated for extended intervals and decline sharply primarily during major stress episodes, such as the early-2020 global liquidity shock and the 2022 deleveraging phase. Importantly, when PTV is above the zero axis (positive domain), bull-state probabilities tend to increase, whereas when PTV falls below zero (negative domain), bull-state probabilities decline and bear-state probabilities rise. This alignment provides descriptive visual evidence that is broadly consistent with the estimated coefficients reported in Table 7: $b_1 > 0$ for bull-state persistence and $b_2 < 0$ for bear-state persistence. In other words, improvements in PTV reinforce regime stability in BTC, while declines in PTV coincide with increased likelihood of bear regimes, consistent with the behavioral predictions of PT.

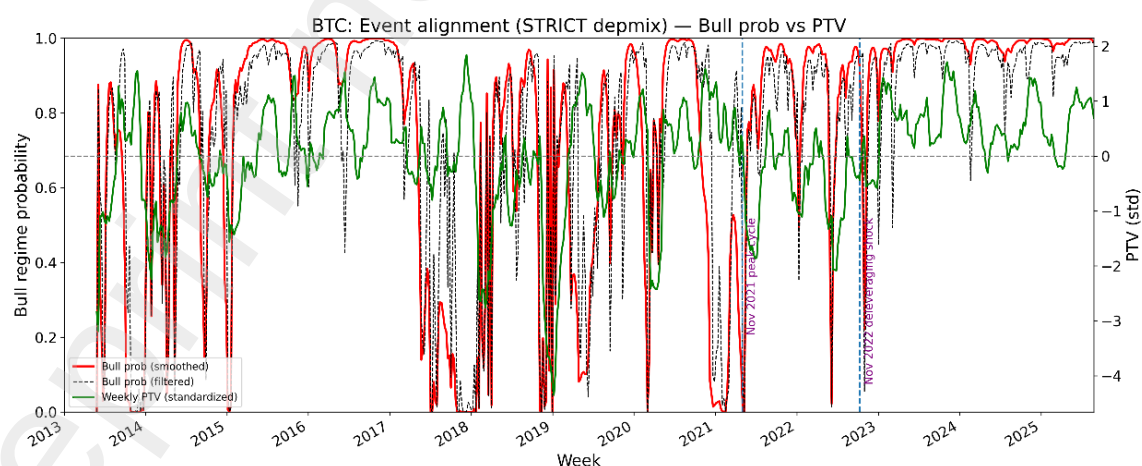


Figure 5. Event-aligned dynamics of bull-state probability and standardized PTV for BTC

For ETH in Figure 6, the bull-state probability path shows frequent jumps and a noticeably less persistent structure compared with BTC. Interestingly, even when weekly PTV is positive, bull probabilities often fail to remain close to unity, and bear probabilities tend to increase. In contrast, negative PTV occasionally coincides with temporary bull probability gains, highlighting the non-linear and somewhat counterintuitive response of ETH to PTV fluctuations. This pattern is consistent with the TVTP-MRS coefficients reported in Table 7 ($b_1 < 0$, $b_2 > 0$), this pattern is consistent with the TVTP-MRS coefficients reported in Table 7, indicating that higher PTV is associated with greater regime instability.

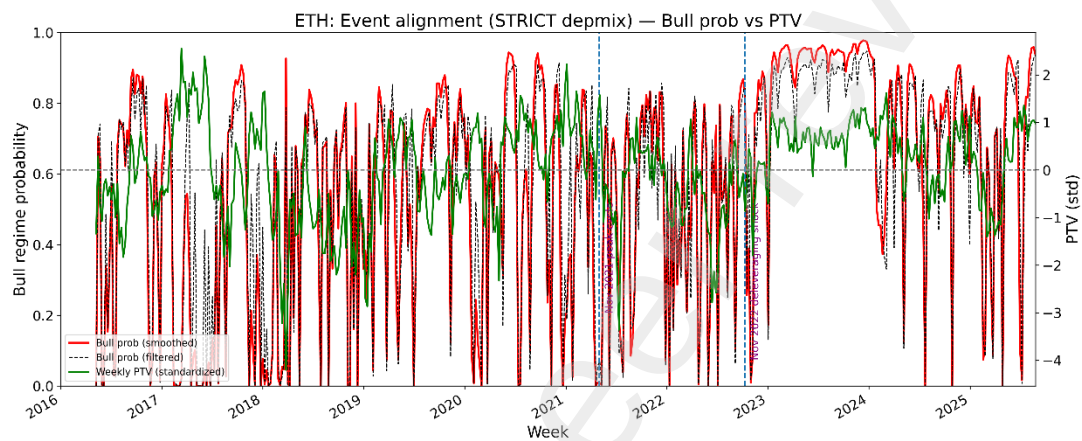


Figure 6. Event-aligned dynamics of bull-state probability and standardized PTV for ETH

A similar dynamic is observed for SOL in Figure 7. Bull-state probabilities experience abrupt changes, particularly during post-2021 episodes such as the 2022 contraction. While PTV occasionally rises above zero, the bull probability does not uniformly increase; rather, bear probabilities often spike, indicating heightened sensitivity to PTV movements. Negative PTV sometimes corresponds to temporary bullish corrections, emphasizing the asymmetry in SOL's regime responses. These observations also align with SOL's transition coefficients in Table 7 and illustrate that, as with ETH, elevated PTV tends to increase regime volatility instead of reinforcing bull phases, though the timing and magnitude of the response are distinct from ETH.

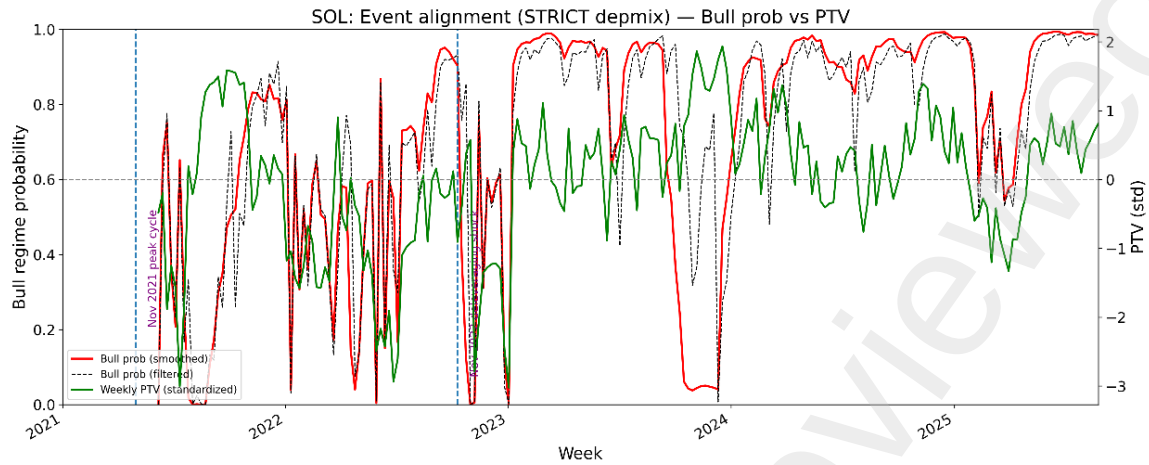


Figure 7. Event-aligned dynamics of bull-state probability and standardized PTV for SOL

4.5 Out-of-sample one-step-ahead forecasting performance

In addition to the in-sample TVTP-MRS analysis, this study evaluates the out-of-sample predictive performance of alternative transition specifications for BTC, ETH, and SOL. The purpose is to examine whether PTV provides complementary predictive information relative to conventional market-state variables. Because forecast accuracy is assessed using observations excluded from parameter estimation, the out-of-sample test provides a stricter evaluation than in-sample coefficient significance.

The effective weekly samples span June 2013 to September 2025 for BTC, May 2016 to September 2025 for ETH, and June 2021 to September 2025 for SOL. To ensure comparison under the same market environment, all models are estimated using observations available up to 25 December 2023 and evaluated from 1 January 2024 to 1 September 2025. This produces 552 estimation observations for BTC, 398 for ETH, and 133 for SOL, with 88 one-step-ahead forecasts for each cryptocurrency. A fixed-window design is adopted: model parameters are estimated once and held fixed throughout the evaluation period, while filtered regime probabilities are updated recursively as realized returns become available. At each time t , the model uses the information available at time t , including the transition variables specified in each competing model. To forecast the return at $t + 1$. The forecasting target can be expressed as:

$$\hat{R}_{t+1|t} = E(R_{t+1}|\mathcal{F}_t) \quad (14)$$

where $\mathcal{F}_t$ denotes the information set available at time t . Specifically, in the two-state TVTP-MRS model, the one-step-ahead forecast is calculated as the weighted average of the expected returns in the two regimes:

$$\hat{R}_{t+1|t} = P(S_{t+1} = 1|\mathcal{F}_t)(\mu_1 + \beta_1 R_t) + P(S_{t+1} = 2|\mathcal{F}_t)(\mu_2 + \beta_2 R_t) \quad (15)$$

where $P(S_{t+1} = 1|\mathcal{F}_t)$ and $P(S_{t+1} = 2|\mathcal{F}_t)$ denote the predicted probabilities of being in state 1 and state 2 at time $t + 1$, respectively. $\mu_i + \beta_i R_t$ represents the conditional expected return in regime i . Therefore, the forecasted return is obtained by weighting the regime-specific expected returns by their corresponding predicted state probabilities.

Forecast accuracy is evaluated using the root mean squared error (RMSE), mean absolute error (MAE), and Theil inequality coefficient (TIC). RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{T} \sum_{t=1}^T (R_{t+1} - \hat{R}_{t+1|t})^2} \quad (16)$$

where R_{t+1} represents the realised next-period return and $\hat{R}_{t+1|t}$ denotes the corresponding one-step-ahead forecast generated at time t . Lower RMSE values indicate better out-of-sample forecasting performance. MAE is defined as:

$$MAE = \frac{1}{T} \sum_{t=1}^T |R_{t+1} - \hat{R}_{t+1|t}| \quad (17)$$

In addition, this study reports the Theil inequality coefficient (TIC) as a scale-adjusted measure of forecast accuracy. TIC is defined as:

$$TIC = \frac{\sqrt{\frac{1}{T} \sum_{t=1}^T (R_{t+1} - \hat{R}_{t+1|t})^2}}{\sqrt{\frac{1}{T} \sum_{t=1}^T R_{t+1}^2 + \frac{1}{T} \sum_{t=1}^T \hat{R}_{t+1|t}^2}} \quad (18)$$

where T denotes the number of one-step-ahead forecasts in the evaluation period. Lower RMSE, MAE, and TIC values indicate better out-of-sample forecasting performance.

To compare PTV with conventional market information, five transition variables are considered. **Return (R_t)** denotes the weekly market return defined previously and captures the overall direction and magnitude of recent market performance. **Downside**

return ($DownR_t$) is defined as

$$DownR_t = \min(R_t, 0), \quad (19)$$

and therefore retains only the negative component of weekly returns, allowing the model to capture the predictive information associated specifically with adverse market movements.

Momentum (MOM_t) measures the persistence of recent price movements and captures short-term trend continuation. It is included because cryptocurrency returns may exhibit momentum-like dynamics, and recent trends may influence both regime persistence and subsequent returns.

Realized volatility (RV_t) measures the magnitude of recent return fluctuations and serves as a proxy for market uncertainty. It is included because changes in volatility may signal shifts in market conditions and therefore contain information relevant to regime-transition and return forecasting.

Unlike these conventional variables, PTV reflects investors' nonlinear psychological evaluation of recent gains and losses relative to a reference point. By incorporating reference dependence, loss aversion, and probability weighting, PTV summarizes the perceived attractiveness of the recent return distribution rather than merely its direction, downside component, trend, or volatility.

Two sets of competing forecasting specifications are estimated. First, the single-variable comparison includes five separate TVTP-MRS models: the R_t -only, $DownR_t$ -only, MOM_t , RV_t -only, and PTV_t -only specifications. In each case, the corresponding variable enters the time-varying transition-probability equations as the sole transition covariate. This comparison evaluates the standalone forecasting performance of each market-state indicator.

Second, an incremental comparison is conducted using a broader conventional benchmark. The conventional benchmark model jointly includes R_t , $DownR_t$, MOM_t , and RV_t in the transition-probability equations. The augmented model then adds PTV to this conventional information set. This comparison examines whether adding PTV improves out-of-sample forecasting performance beyond the combined information contained in conventional return, downside, momentum, and volatility variables. Table 9 confirms that Panel A reports the five single-variable specifications, while Panel B compares the four-variable conventional benchmark with its PTV-augmented counterpart.

The purpose of this exercise is not to claim that PTV necessarily dominates every

conventional predictor across all cryptocurrencies and evaluation measures. Rather, it assesses whether PTV provides useful complementary forecasting information.

Table 9. Out-of-sample forecasting performance for weekly cryptocurrencies

Panel A: Single-variable specifications					
Market	Model	PTV included	RMSE	MAE	TIC
BTC	R_t -only	No	0.0572	0.0438	0.8688
BTC	$DownR_t$ -only	No	0.0572	0.0438	0.8696
BTC	MOM_t -only	No	0.0571	0.0438	0.8658
BTC	RV_t -only	No	0.0571	0.0438	0.8681
BTC	PTV_t -only	Yes	0.0570	0.0438	0.8656
ETH	R_t -only	No	0.1001	0.0770	0.8744
ETH	$DownR_t$ -only	No	0.1000	0.0770	0.8742
ETH	MOM_t -only	No	0.0996	0.0765	0.8903
ETH	RV_t -only	No	0.0995	0.0765	0.8815
ETH	PTV_t -only	Yes	0.1007	0.0776	0.8796
SOL	R_t -only	No	0.1244	0.0969	0.8617
SOL	$DownR_t$ -only	No	0.1252	0.0974	0.8638
SOL	MOM_t -only	No	0.1245	0.0972	0.8266
SOL	RV_t -only	No	0.1236	0.0956	0.8610
SOL	PTV_t -only	Yes	0.1236	0.0956	0.8576
Panel B: Incremental comparison based on R_t, $DownR_t$, MOM_t, and RV_t					
Market	Model	PTV included	RMSE	MAE	TIC
BTC	Conventional benchmark	No	0.0573	0.0437	0.8748
BTC	Conventional benchmark + PTV_t	Yes	0.0572	0.0438	0.8692
ETH	Conventional benchmark	No	0.0995	0.0764	0.8851
ETH	Conventional benchmark + PTV_t	Yes	0.0996	0.0766	0.8890
SOL	Conventional benchmark	No	0.1260	0.0980	0.8004
SOL	Conventional benchmark + PTV_t	Yes	0.1267	0.0977	0.8431

Notes: Panel A reports single-variable TVTP-MRS forecasting specifications. In Panel B, the conventional benchmark includes R_t , $DownR_t$, MOM_t , and RV_t , whereas the augmented specification additionally includes PTV_t . All information available in week t is used to forecast the return in week $t + 1$. Lower RMSE, MAE, and TIC values indicate better out-of-sample forecasting performance.

Table 9 suggests that PTV may provide complementary forecasting information beyond conventional market-state variables. In Panel A, the PTV_t -only specification achieves the lowest RMSE and TIC for BTC and the lowest RMSE and MAE for SOL, indicating that PTV captures predictive features that may not be fully reflected in return, downside return, momentum, or volatility. Panel B provides further evidence that the inclusion of PTV improves selected forecasting measures, including RMSE and TIC for BTC and MAE for SOL. Taken together, these results indicate that PTV can complement conventional predictors, with its forecasting contribution varying across cryptocurrencies and evaluation measures.

4.6 Robustness and Supplementary Analyses

4.6.1 MRS model

To assess the robustness of the underlying regime-dependent return structure identified by the main TVTP-MRS model, this study re-estimates the model using a standard constant-transition MRS specification. Unlike the TVTP specification, in which transition probabilities vary with behavioral inputs such as PTV, the MRS model assumes constant transition probabilities and excludes external explanatory variables. This simplified setup allows for a more controlled assessment of whether the regime-dependent return dynamics identified in the main model remain consistent under a constrained setting. The model is specified as follows: Formula 20.

$$R_t = \mu_{S_t} + \beta_{S_t} R_{t-1} + \sigma_{S_t} \varepsilon_t \quad \varepsilon_t \sim N(0,1) \quad (20)$$

where μ_{S_t} , β_{S_t} , and σ_{S_t} represent the regime-specific intercept, autoregressive coefficient, and standard deviation of the error term, respectively. Estimation results for the weekly cryptocurrencies sample are reported in Figure 8 and Table 10.

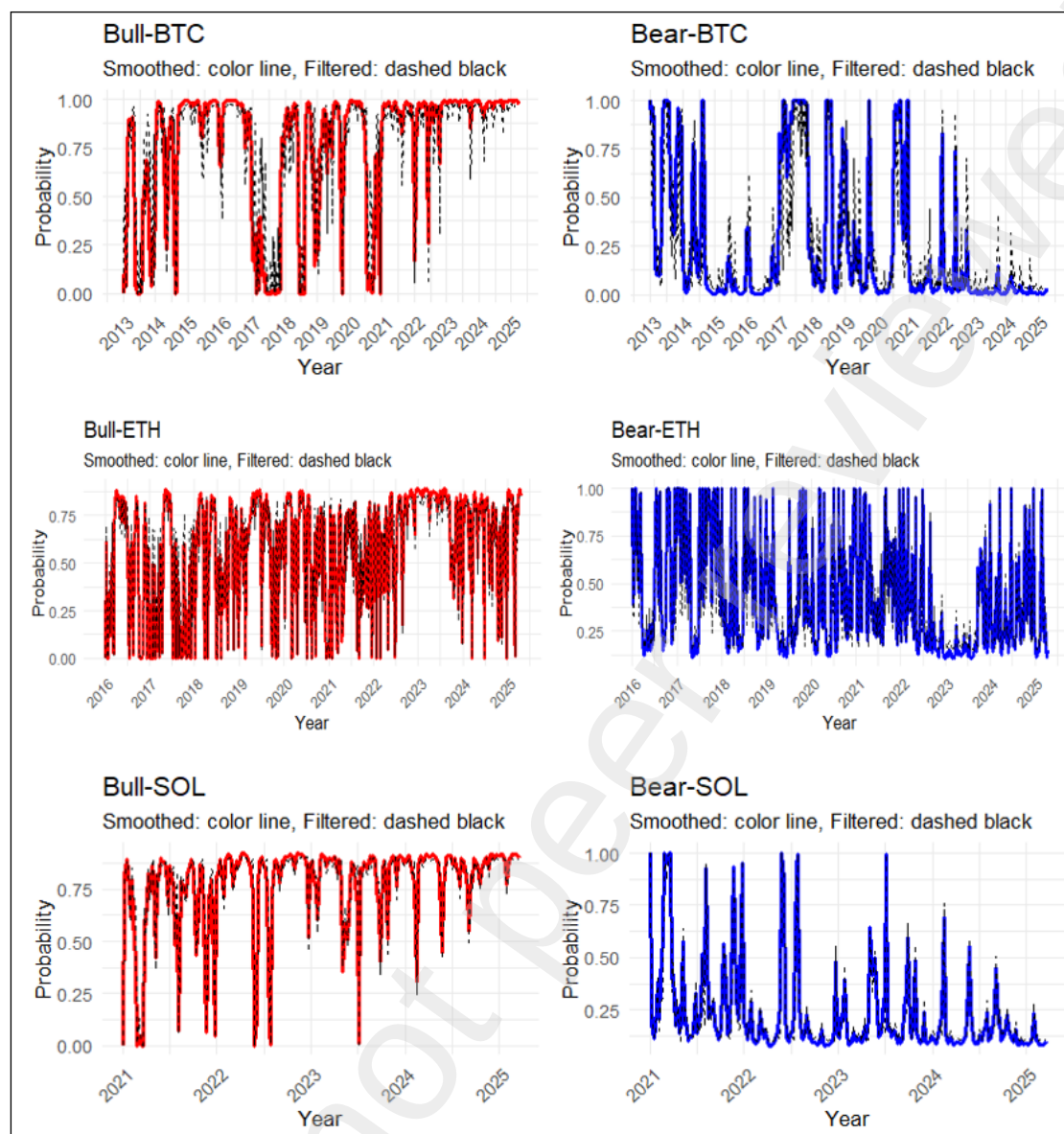


Figure 8. Filter & Smoothed probabilities of the MRS model of weekly cryptocurrencies

Table 10. MRS test results of weekly cryptocurrencies

<i>Panel A: Estimated sample coefficients</i>						
Markets	μ_1	μ_2	β_1	β_2	σ_1	σ_2
BTC	0.007** (0.042)	0.005 (0.728)	0.064 (0.180)	0.216*** (0.010)	0.070*** (0.000)	0.174*** (0.000)
ETH	0.007 (0.194)	0.015 (0.289)	-0.121** (0.023)	0.165* (0.086)	0.065*** (0.000)	0.195*** (0.000)
SOL	0.004 (0.699)	0.007 (0.869)	-0.156 (0.127)	0.424* (0.069)	0.115*** (0.000)	0.238*** (0.000)
<i>Panel B: Transition probabilities and durations</i>						
Markets	P_{11}	P_{22}	P_{12}	P_{21}	DU ₁	DU ₂
BTC	0.962*** (0.000)	0.874*** (0.000)	0.038** (0.035)	0.126*** (0.010)	26.26 weeks	7.94 weeks
ETH	0.710*** (0.000)	0.612*** (0.000)	0.290*** (0.000)	0.388*** (0.000)	3.45 weeks	2.58 weeks
SOL	0.850*** (0.000)	0.479** (0.024)	0.150 (0.194)	0.521** (0.014)	6.67 weeks	1.92 weeks

Notes: *, **, and *** denote significance at 10%, 5%, and 1% level. The values in parentheses represent p-values.

The constant-transition MRS results in Table 10 broadly preserve the two-state structure identified by the TVTP-MRS model. Across all three cryptocurrencies, the first regime is characterized by lower volatility and a longer expected duration, whereas the second regime exhibits higher volatility and shorter persistence. This pattern is particularly pronounced for BTC, for which the expected durations of the two regimes are 26.26 and 7.94 weeks, respectively. The similarity in regime-specific means, autoregressive coefficients, volatilities, and state-duration patterns indicates that the underlying bull–bear structure is not mechanically generated by allowing PTV to enter the transition equations. However, because the standard MRS model excludes PTV, this exercise validates the robustness of the underlying regime structure rather than the robustness of the estimated PTV coefficients.

4.6.2 Using the TVTP-MRS model in the U.S. stock market

We further examine whether the BTC-style sign structure is observed in a more mature external market environment. To provide a supplementary external-market comparison, we examine whether a similar directional pattern is observed in three major U.S. stock indices: the S&P 500, NASDAQ Composite, and Dow Jones Industrial Average, using data obtained from the CRSP (Center for Research in Security Prices) database. The U.S. robustness checks are conducted over the period from January 2006 to December 2023, covering key market episodes such as the Global Financial Crisis, the European debt crisis, and the COVID-19 pandemic. Figure 9 presents the smoothed and filtered regime probabilities for these indices, visually confirming that the model captures economically interpretable bull–bear dynamics throughout the sample period, particularly during pronounced stress events such as the 2008 financial crisis and the

COVID-19 recession.

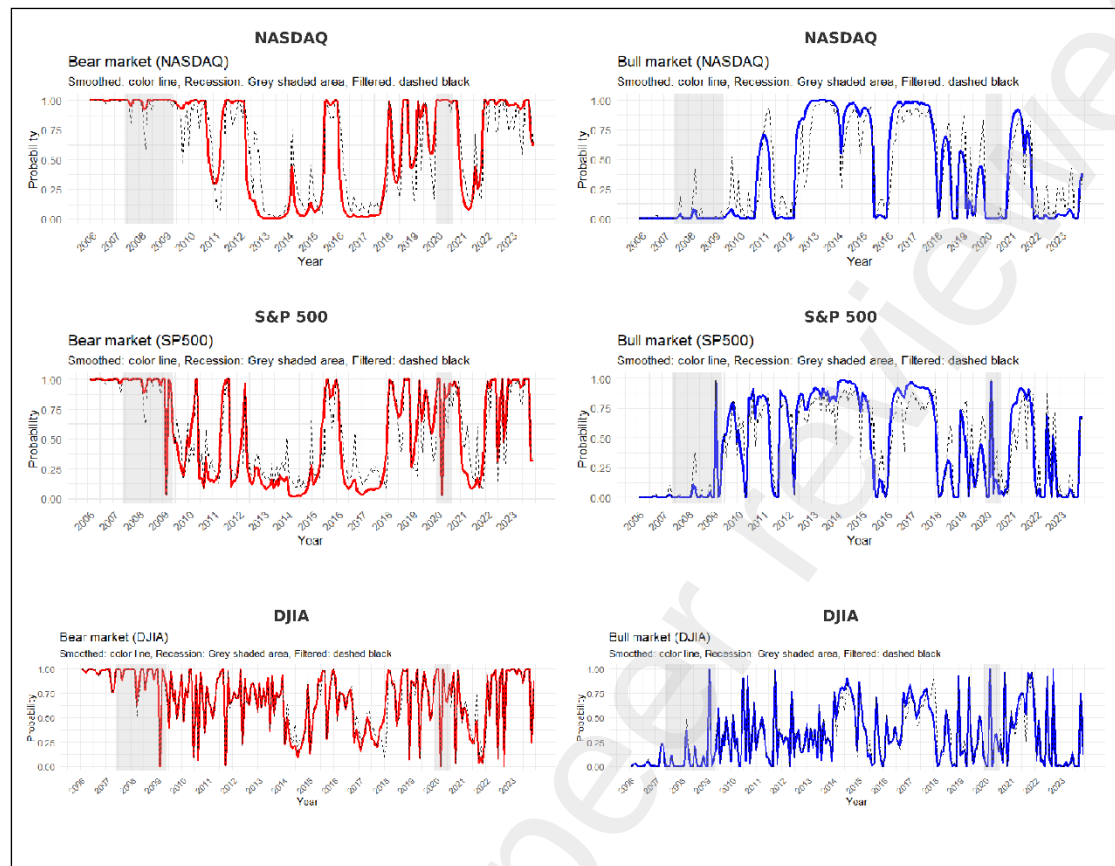


Figure 9. Filter & Smoothed probabilities of U.S. stock indices

The corresponding parameter estimates are reported in Table 11. Although some coefficients exhibit directional similarity to the BTC results, they are statistically insignificant and therefore do not provide strong confirmatory evidence.

Table 11. TVTP-MRS model results for the U.S. stock indices

<i>Panel A: Estimated sample coefficients</i>						
Shares	μ_1	μ_2	β_1	β_2	σ_1	σ_2
NASDAQ	0.021*** (0.000)	-0.013** (0.034)	-0.460*** (0.000)	0.166* (0.057)	0.023*** (0.000)	0.064*** (0.000)
S&P 500	0.021*** (0.000)	-0.021*** (0.001)	-0.459*** (0.000)	0.322** (0.020)	0.025*** (0.000)	0.049*** (0.000)
DJIA	0.021*** (0.002)	-0.018*** (0.000)	-0.524*** (0.000)	0.421*** (0.000)	0.027*** (0.000)	0.040*** (0.000)
<i>Panel B: Transition probabilities and estimated transition coefficients</i>						
Shares	a ₁	a ₂	b ₁	b ₂		
NASDAQ	-2.075*** (0.003)	2.310*** (0.008)	24.800 (0.298)	-16.484 (0.407)		
S&P 500	-2.402* (0.076)	1.852** (0.019)	9.045 (0.491)	-17.707 (0.435)		
DJIA	0.700 (0.466)	-1.260** (0.027)	78.085 (0.211)	-5.143 (0.704)		

Notes: *, **, and *** denote significance at 10%, 5%, and 1% level. The values in parentheses represent p-values.

The results for the three major U.S. stock indices reveal a notable directional correspondence with the BTC estimates. Across the NASDAQ, S&P 500, and DJIA, the PTV coefficient associated with the persistence of the first regime is positive, whereas that associated with the persistence of the second regime is negative. This common sign pattern is consistent with the possibility that a more favorable PTV strengthens the persistence of favorable market conditions while reducing the persistence of unfavorable conditions across different asset classes. The corresponding coefficients, however, are not statistically significant in the U.S. market. Accordingly, the U.S. evidence is more appropriately interpreted as directional cross-market consistency than as a statistical replication of the BTC results.

This difference in statistical strength also provides insight into the potential boundary conditions of the PTV effect. Cryptocurrency markets are characterized by weaker fundamental valuation anchors, greater retail participation, higher volatility, and stronger sensitivity to speculative narratives and investor sentiment. These features may allow PT perceptions of gains and losses to play a more visible role in regime-transition dynamics. By contrast, in the deeper and more institutionally developed U.S. equity market, the influence of PTV may be diluted by a broader set of fundamental, macroeconomic, and institutional forces. The weaker U.S. estimates are therefore consistent with the interpretation that the relevance of PTV is more pronounced in markets where prices are more strongly shaped by sentiment and subjective valuation. Nevertheless, because the U.S. coefficients lack statistical significance, this market-structure interpretation should be regarded as suggestive rather than conclusive. Overall, the U.S. results preserve the directional pattern observed for BTC while indicating that

the magnitude and econometric detectability of the PTV effect depend on the characteristics of the market in which it operates.

5. Practical Implications and Limitations

5.1 Practical implications

The findings of this study offer several practical implications for investors, risk managers, exchanges, and regulators operating in cryptocurrency markets. First, the results indicate that PTV contains information relevant not only to static pricing outcomes but also to the persistence of market regimes. In particular, the BTC estimates show that higher PTV increases bull-regime persistence, whereas lower PTV reinforces bear-regime persistence. This suggests that PTV may serve as a useful behavioral state-conditioning indicator for market participants seeking to assess whether favorable or adverse market conditions are likely to persist. Rather than treating behavioral valuation merely as a contemporaneous correlate of price movements, investors may use PTV as an additional signal when evaluating whether market sentiment is becoming sufficiently strong to sustain a bull phase or sufficiently weak to prolong a bear phase.

Second, the results carry important implications for risk management and market monitoring. Because the transition effects of PTV are asymmetric, especially in the BTC market, deteriorating behavioral valuation may signal not only weaker sentiment but also a higher likelihood that unfavorable regimes will persist. In highly volatile digital asset markets, this implies that conventional risk indicators such as realized volatility or drawdown measures may be complemented by behavioral state variables such as PTV. In practical terms, PTV may be incorporated into regime-dependent stress testing, tactical rebalancing decisions, and early-warning systems designed to detect periods in which downside market conditions are becoming more persistent.

Third, the clear heterogeneity across BTC, ETH, and SOL suggests that behavioral indicators should not be transferred mechanically across cryptocurrencies. BTC exhibits the most coherent and interpretable sign structure, whereas ETH and SOL exhibit a sign pattern that differs from BTC, although the statistical strength is weaker and less uniform, especially for ETH. This implies that the regime effect of behavioral valuation depends on asset-specific characteristics, including market maturity, ecosystem structure, narrative sensitivity, and investor composition. For practitioners, the implication is that a single behavioral trading rule is unlikely to perform uniformly across all digital assets. For researchers and supervisors of digital asset markets, the broader implication is that cryptocurrency markets should not be treated as behaviorally homogeneous.

Fourth, the robustness exercise based on the three major U.S. stock indices provides an additional comparative implication. Although the PTV-related coefficient pattern in the U.S. market is directionally consistent with the BTC results, its weaker statistical strength suggests that the measurable effect of behavioral valuation declines as market depth and institutional sophistication increase. This indicates that sentiment-

linked behavioral indicators such as PTV may be especially informative in retail-driven and narrative-sensitive markets, while their direct empirical salience becomes weaker in more mature financial systems where a broader set of macro-financial forces shapes regime transitions.

5.2 Limitations

Despite these contributions, this study has several limitations that should be acknowledged. First, the empirical analysis focuses on only three major cryptocurrencies. Although BTC, ETH, and SOL provide meaningful variation in market maturity, investor composition, and ecosystem structure, the findings may not be generalizable to the broader universe of digital assets, especially smaller tokens, stablecoins, or cryptocurrencies with very short and discontinuous trading histories. Future research may extend the present framework to a larger cross-section of digital assets to assess whether the documented transition effects of PTV remain stable across different segments of the cryptocurrency market. In addition, the SOL results should be interpreted with some caution because, after accounting for the rolling PTV window, weekly aggregation, and model lag structure, its effective sample contains 221 weekly observations, substantially fewer than those for BTC and ETH. This shorter sample may reduce the precision of the TVTP-MRS estimates.

Second, the construction of PTV relies on a specific rolling-window design and the benchmark cumulative PT parameterization. This approach is consistent with the established PT literature and facilitates comparability with prior studies, but it also imposes modelling choices regarding memory horizon, reference-point treatment, and probability weighting. Alternative window lengths, reference-point definitions, or parameter calibrations may affect the estimated magnitude of the transition effects. Future research may therefore further examine the robustness of the findings under alternative CPT parameterizations, including those proposed by Abdellaoui (2000), Andersen et al. (2006), and Harrison and Rutström (2009), as well as alternative rolling windows and reference-point definitions.

Finally, the weekly estimation frequency improves regime interpretability in highly volatile cryptocurrency markets, but it may also smooth over shorter-lived behavioral adjustments that unfold at the daily or intraday level. Since digital asset markets trade continuously and often react quickly to sentiment and narrative shocks, some relevant behavioral responses may occur at a faster horizon than the one examined here. Future research may therefore explore whether similar asymmetries emerge in daily or mixed-frequency regime-switching specifications.

6. Conclusion

This study examines whether PTV helps explain regime persistence in cryptocurrency markets. By embedding PTV in a two-state TVTP-MRS framework for BTC, ETH, and SOL, the analysis moves beyond the usual question of whether behavioral valuation predicts returns and instead asks whether it helps account for the continuation of bull

and bear market states.

The results show that PTV is not merely a static behavioral correlate of cryptocurrency returns. It enters the transition process itself. More importantly, it does not do so in a uniform way. In BTC, the estimated coefficient pattern is economically intuitive and internally consistent: higher PTV is associated with stronger bull-regime persistence, whereas lower PTV reinforces bear-regime persistence. This suggests that, in the most mature and macro-sensitive cryptocurrency in the sample, PTV operates through a coherent state-dependent channel. When recent return prospects become behaviorally more attractive, optimistic market conditions are more likely to persist; when behavioral valuation deteriorates, adverse states become harder to escape.

That mechanism does not generalize mechanically across assets. ETH and SOL exhibit the opposite coefficient pattern, indicating that the role of behavioral valuation in regime dynamics is conditional on market structure rather than universal across cryptocurrencies. This difference is important. It suggests that the effect of PTV depends on how each asset is traded, narrated, and interpreted by market participants. In that sense, the findings do not simply show that sentiment matters in cryptocurrency markets. They show that the same behavioral signal can enter regime persistence differently across digital assets with different levels of maturity, ecosystem dependence, and investor composition.

The supplementary analyses provide contextual evidence that is broadly consistent with this interpretation, although they do not constitute formal confirmation of the proposed mechanism. This interpretation is most clearly supported for BTC. For ETH and SOL, the estimated sign patterns are weaker and less uniform and should be interpreted as asset-specific associations rather than evidence of a common bull–bear transition mechanism. The U.S. stock market extension points in the same direction: the sign structure remains broadly consistent, but the statistical strength becomes weaker in deeper and more institutionally dominated markets. Taken together, these results suggest that the relevance of PTV is not confined to one specification, but its empirical salience depends on the market environment in which it operates.

The broader implication of this study is that PT can be brought into the analysis of market-state dynamics in a way that is empirically tractable and economically meaningful. For the behavioral finance literature, the paper shows that PTV matters not only for pricing outcomes but also for the persistence of market regimes. The cryptocurrency literature shows that regime dynamics are shaped by investor psychology in ways that are both asymmetric and asset-specific. More generally, the study provides a framework for linking behaviorally perceived return distributions to cyclical risk in highly volatile markets. Future research may extend this approach to a wider set of digital assets, alternative behavioral proxies, and richer identification strategies for the channels through which sentiment affects regime persistence.

Declarations:

The authors declare no conflict of interest concerning the results and content.

Data availability:

The data is available on request.

Conflict of Interest declaration: The authors declare that they have no affiliations with or involvement in any organization or entity with any financial interest in the subject matter or materials discussed in this manuscript.

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